

**VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY**



VU MINH TUAN

**JOINT EVENT EXTRACTION
WITH DISTANT SUPERVISION
AND DEPENDENCY-AWARE LEARNING**

STUDENT SCIENTIFIC RESEARCH REPORT

Faculty of Information Technology

HANOI - 2025

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Abstract

Event extraction has always been a field of interest for researchers, encompassing four main tasks: Event Mention Detection, Event Detection, Relation Extraction, and Event Argument Extraction. Since the 1990s, various methods have been proposed to tackle these tasks, ranging from traditional approaches like Pattern Matching to more advanced Machine Learning techniques. In recent years, with the rapid advancements in deep learning, numerous studies have explored deep learning-based approaches for event extraction. However, most of these approaches have primarily focused on widely used languages such as English and Chinese. Unfortunately, for Vietnamese, there is only one publicly available dataset for Event Extraction, covering three main tasks: Event Mention Detection, Event Detection, and Event Argument Extraction. Moreover, experiments conducted on this dataset have not yielded high-performance results, highlighting the challenges in Vietnamese event extraction. To address this limitation, we introduce a Semi-supervised method, specifically pseudo-labeling, based on event occurrence frequency, combined with a Schema-based data augmentation approach. This method aims to enhance both the quantity and quality of training data. Our experimental results demonstrate that the enriched dataset is more diverse and leads to improved performance, achieving a higher F1 score compared to the original dataset.

Acknowledgements

I would like to thank my supervisor Dr. Tran Mai Vu of the Data Science and Knowledge Technology Laboratory at VNU University of Engineering and Technology. He consistently allowed this report to be my own work, but steered me in the right the direction whenever he thought I needed it.

Although I have tried to complete the report but will inevitably have small mistakes. I respectfully receive the understanding and guidance from the teachers and the jury.

Declaration

I declare that the report has been composed by myself and that the work has not be submitted for any other degree or professional qualification. I confirm that the work submitted is my own, except where work which has formed part of jointly-authored publications has been included. My contribution and those of the other authors to this work have been explicitly indicated below. I confirm that appropriate credit has been given within this report where reference has been made to the work of others.

I certify that, to the best of my knowledge, my report does not infringe upon anyone's copyright nor violate any proprietary rights and that any ideas, techniques, quotations, or any other material from the work of other people included in my report, published or otherwise, are fully acknowledged in accordance with the standard referencing practices.

I take full responsibility and take all prescribed disciplinary actions for my commitment.

Student

Vu Minh Tuan

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Acronyms

BERT Bidirectional Encoder Representations from Transformers

EAE Event Argument Extraction

ED Event Detection

EE Event Extraction

EMD Entity Mention Detection

GCN Graph Convolutional Networks

GPT Generative Pre-training Transformer

IE Information Extraction

LLM Large Language Models

NLP Natural Language Processing

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Chapter 1: Introduction

1.1. Motivation

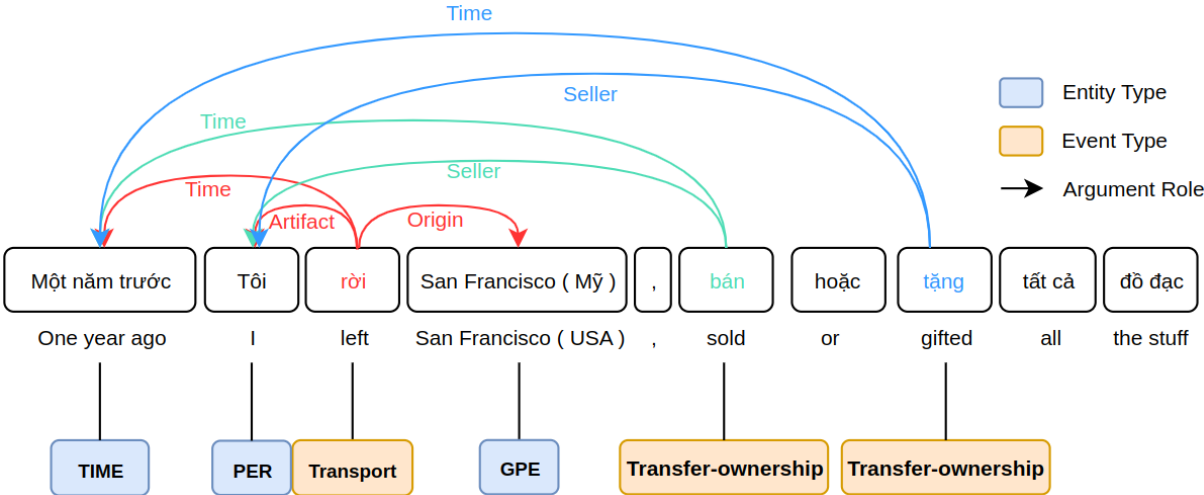


Figure 1.1: An example of Event Extraction

Event Extraction (EE) is a fundamental yet complex task in Information Extraction (IE), consisting of four sub-tasks: Entity Mention Detection (EMD), Event Detection (ED), Relation Extraction (RE) and Event Argument Extraction (EAE) [1]. To better understand these problems consider the example in Figure 1.1. Event Extraction aims at identifying and structuring event-related information from unstructured text. It plays a crucial role in natural language understanding by detecting events, their participants, and contextual details. An event is an occurrence that takes place at a specific time and location, involving one or more entities, often characterized by a change in state. Event extraction focuses on identifying events within text and, when found, determining their type, participants, and relevant details. While event types may vary in structure, the main objective is to transform unstructured text into a structured format. This allows for better understanding of real-world events by answering key questions such as “who, what, when, where, why, and how” using multiple sources (e.g. news articles, social media).

Event extraction has a wide range of applications across different fields [21]. One of its most popular uses is in building structured event representations that can enhance knowledge bases [15]. Event detection and monitoring have long been used to man-

age public affairs, as staying informed about emerging social events allows for timely responses [3, 16]. In the biomedical field, it plays a role in identifying changes in biomolecules (such as genes and proteins) or interactions between them, which are often described in scientific literature to better understand physiological and disease mechanisms [17]. Overall, event extraction techniques offer valuable benefits across multiple domains.

Although event extraction has many promising applications, it remains a challenging task due to the complexity of event structures and their diverse components. Various methods have been proposed to deal with the Event Extraction tasks [21]. In recent years, Deep Learning approaches have gained significant popularity. However, one of the biggest challenges in event extraction using deep learning, particularly for low-resource languages like Vietnamese, is the lack of high-quality labeled data. Unlike English and Chinese, which have well-established event extraction datasets such as ACE 2005 [20] and RAMS [7], Vietnamese event extraction research has been severely limited by the absence of large-scale annotated corpora. The BKEE dataset, while being the first Vietnamese dataset for event extraction, is relatively small and lacks sufficient diversity to cover the wide range of event structures encountered in real-world scenarios. Manually annotating new datasets is expensive, time-consuming, and requires domain experts to ensure high-quality labeling. To address this limitation, distant supervision provides a viable alternative by automatically generating labeled data from unlabeled text. While distant supervision can generate large amounts of labeled data without human annotation, it introduces label noise, as not all extracted sentences truly contain the intended event. Therefore, an effective filtering mechanism is crucial to improving the quality of pseudo-labeled data. In this work, we employ event occurrence frequency filtering to improve distant supervision. By leveraging N-ary relations—which analyze the frequency of event mentions and their associated entities across multiple articles covering the same topic—we can filter out unreliable pseudo-labels and retain only high-confidence training data.

However, using a pipeline approach like traditional event extraction systems, where each task—Entity Mention Detection (EMD), Event Detection (ED), and Event Argument Extraction (EAE)—is handled by a separate model in a sequential manner, is not ideal. While this modular approach allows for independent optimization of each task, constructing multiple models is not resource-efficient, and it also suffers from a critical weakness: error propagation [21]. Mistakes made in earlier stages (e.g., incorrectly detecting an entity or event trigger) directly impact downstream tasks, leading to cas-

cading errors that degrade overall performance. For instance, if an entity mention is misclassified in the EMD stage, it affects both the ED and EAE stages, as these rely on correctly identified entities to extract event triggers and arguments. Additionally, pipeline models fail to leverage cross-task dependencies, treating each task as independent rather than recognizing their inherent relationships. This limitation prevents the model from utilizing global context effectively, which is crucial for accurately predicting events and arguments. To overcome these issues, we adopt a joint model that simultaneously predicts entities, event triggers, and arguments. Furthermore, our joint model incorporates dependency-aware learning through graph-based representations, explicitly modeling the relationships between event components. By using Graph Convolutional Networks (GCNs) to structure instance interactions and type dependencies, we further enhance the model’s ability to learn from contextual information.

1.2. Problem Statement

Event extraction (EE) is the process of identifying event triggers, classifying event types, and extracting related entities with their roles from unstructured text to create structured representations of real-world events. As aforementioned, Event Extraction comprises 4 sub-tasks: EMD, ED, RE, and EAE [1]. In this study, we aim to extract events from individual sentences, referred to as sentence-level extraction, rather than document-level extraction.

Generally, the goal of Entity Mention Detection (EMD) is to identify and classify words or phrases referring to real-world entities, such as people, organizations, locations, and dates. The goal of Event Detection (ED) is to identify event triggers, which are specific words or phrases that indicate the occurrence of an event (e.g., "acquired" in a merger event or "resigned" in a resignation event). Each trigger is categorized into predefined event types to facilitate structured event representation. Relation Extraction (RE) plays a crucial role in structuring information by identifying and classifying relationships between entities within a text. Finally, Event Argument Extraction (EAE) aims to determine which entities are involved in an event and assigns them specific roles (e.g., "buyer" in a purchase event or "location" in a disaster event). However, in this study, we only perform EMD, ED, and EAE due to the characteristics of the BKEE dataset [14]. There are a total of 12 entity types and 8 event types, which are further divided into 33 sub-event types, along with 28 argument roles [14]. In this study, we treat sub-event types as event types. The types of entities, event triggers, and argument roles are labeled

based on these type and role sets.

We define the task as follows:

- The input is a plain sentence in natural language S .
- The output is a structured representation containing the detected entities, event triggers, and arguments. Each entity, event trigger, and argument is associated with an entity type, event type, and argument role, respectively. In this context, the argument arg_{ij} represents the relation between the event trigger t_i and entity e_j .

1.3. Difficulties and Challenges

Event extraction in Vietnamese presents several unique challenges due to linguistic, resource-related, and computational limitations. These are the key challenges that we faced:

- **Lack of clear word boundaries:** Unlike English, where words are typically separated by spaces, Vietnamese is an isolating language, meaning words are formed from sequences of syllables (e.g., *học sinh* for "student"). Tokenization errors can impact event extraction performance since incorrect segmentation may change the meaning of a phrase.
- **Ambiguity in word sense and structure:** Vietnamese relies heavily on context to determine meaning. A single word or phrase can have multiple interpretations. For example, *bán nhà* can mean "selling a house" (*bán* = "sell", *nhà* = "house") but could also refer to a person's last name (*Bán* = surname, *Nhà* = noun classifier). Extracting the correct event type in such cases requires deep contextual understanding.
- **Scarcity of benchmark datasets:** Compared to English datasets like ACE 2005, RAMS, or MAVEN, Vietnamese has no event extraction datasets, except BKEE. The BKEE dataset is one of the first Vietnamese event extraction corpora, but it is still small and lacks sufficient diversity in event types and argument structures.
- **Ambiguous named entities:** Many Vietnamese names, locations, and organizations overlap with common words. For example, *Bình* can be a person's name, a province, or mean "peace." Without context, an NER model may misclassify such entities.

- **Polysemous words as event triggers:** Many Vietnamese words have multiple meanings depending on context. For example, *chạy* can mean "run" in a physical sense, but in *chạy chức*, it means corruption. Extracting the right event type requires deep semantic understanding.
- **Multi-word triggers:** Unlike English, where many triggers are single words (e.g., attack, resign), Vietnamese event triggers often involve multiple words. For example, *tuyên bố từ chức* ("announced resignation") consists of multiple words that must be recognized together as a single event trigger.

1.4. Contributions of the Research

Currently, Vietnamese Event Extraction remains a challenging task and has yet to be solved effectively. One of the biggest hurdles is the scarcity of high-quality datasets that are both rich and diverse, carefully annotated by experts. Creating such a dataset is undeniably expensive, time-consuming, and requires significant human effort. Because of this, even the ACE 2005 dataset costs thousands of dollars for its event extraction annotations. Another major challenge is building an efficient model that can accurately recognize Vietnamese entities, event triggers, and arguments, given the unique characteristics of the Vietnamese language, as mentioned earlier.

Due to these limitations, an effective method is needed to automatically generate well-annotated data and use it for model training. With these challenges as motivation, our main contributions are as follows:

- **In terms of materials:** We propose an enhanced distant supervision method, which will leverage the unlabeled data to generate pseudo-labelled data, subsequently filtering based on the occurrences frequency of the events. Because of the characteristics of Vietnamese news articles across sites, in which the same events across articles of the same topic can be mentioned concurrently, this technique can ensure the selected data is rightly predicted, filtering out the sentences in which events do not occur frequently. Moreover, a schema-based data augmentation method is used to enhance the diversity of the data, thus, the model can predict better in many different situations, improving the performance of the model.
- **In terms of methodological:** We use a joint model to predict entities, events, and arguments simultaneously, rather than a pipeline approach where each task is handled by a separate model, and each step depends on the output of the previous

one. The pipeline approach can lead to error propagation, where mistakes in earlier stages negatively impact later ones. In contrast, a joint model processes all components at the same time, enabling better information sharing and reducing reliance on potentially incorrect intermediate results. Additionally, the joint model leverages global context, allowing it to capture complex interactions between entities, triggers, and arguments more effectively.

- **In terms of result:** Compared to the results reported by Nguyen *et al.* [14], our best F1 score outperforms theirs in all three tasks: EMD, ED, and EAE.

1.5. Structure of the Report

This scientific research report includes five main Chapters and one Conclusion, as follows:

Chapter 1: Introduction. This Chapter is an introduction to the tasks of Vietnamese Event Extraction. We present the motivations, problem statement, key challenges, and contributions of this research.

Chapter 2: Related Works. This chapter presents related work relevant to the task. It covers the methods and development of distant supervision, data augmentation, and joint models for event extraction.

Chapter 3: Material This chapter discusses the materials covered in this report. We provide detailed methods for generating high-quality annotated data and enhancing data diversity through data augmentation.

Chapter 4: Methods This chapter discusses the methods this report is focused on. We will provide the detailed methods of our joint model for Event Extraction.

Chapter 5: Experiments and Results. This chapter analyzes the performance of our joint model on the new dataset. Additionally, we highlight the impact of each proposed method on the overall performance of the model.

Conclusions. This Chapter concludes the report by summarizing the important contributions and results. Also, we highlight the limitations of our methods and point out some further extensions in future work.

Chapter 2: Related Works

Previous studies on generating annotated data for Event Extraction primarily focus on one of four tasks: Entity Mention Detection, Event Detection, Relation Extraction, or Event Argument Extraction [2, 8, 9, 12]. Most of these works explore Distant Supervision or Semi-Supervised Learning to create labeled data. Additionally, data augmentation techniques have been proposed to enhance data diversity and improve model robustness. With the rapid advancement of deep learning, joint models leveraging neural networks have become widely adopted for tackling all four Event Extraction tasks simultaneously. In this section, we discuss previous research on automatic labeled data generation and Event Extraction using joint models with dependency-aware learning.

2.1. Model For Event Extraction

In the 1990s and 2000s, several Pattern Matching methods were introduced, which can be reduced to two main ideas: Manual Pattern Construction and Automatic Pattern Construction [21]. Pattern matching methods first define specific event templates based on predefined rules or patterns. Once these templates are constructed, the system identifies relevant events in text by matching them against these patterns. This approach relies on linguistic structures, keywords, and syntactic cues to extract event information accurately. Following this, Machine Learning approaches emerged, relying on algorithms such as Support Vector Machines (SVM), Maximum Entropy (ME), etc., as well as Feature Engineering techniques [21]. In recent years, Deep Learning approaches have gained significant popularity. Deep learning approaches leverage neural networks (Recurrent Neural Network, Convolutional Neural Network, etc.) to accomplish Event Extraction tasks [21].

There have been several deep learning approaches that handle EE tasks simultaneously instead of constructing a pipeline with multiple models to avoid error propagation, such as the framework DyGIE++ proposed by Wadden *et al.* [19], and the framework OneIE proposed by Lin *et al.* [11]. However, there are some flaws of these two models that need to be improved. These works suffer from limitations in capturing cross-task and cross-instance dependencies. Wadden *et al.* rely on shared hidden layers but do not explicitly model interactions between related instances, leading to suboptimal rep-

resentation learning [13]. OneIE introduces global type dependencies but only apply them during decoding, preventing these constraints from contributing to representation learning during training [13].

The model FourIE explicitly models inter-task dependencies to improve performance [13]. It achieves this through two key innovations: an instance interaction graph and a type dependency graph. The model’s architecture comprises three main components. First, a span detection module identifies spans corresponding to entity mentions and event triggers using a BERT encoder followed by two Conditional Random Fields (CRFs). Second, the instance interaction module constructs and processes the interaction graph to refine instance representations before predicting labels. Third, the type dependency-based regularization module builds and utilizes type dependency graphs to constrain predictions and inject additional structural knowledge into the training process.

Applied in our approach: We use this model to train on the generated dataset because it is language-independent, making it suitable for various languages, including Vietnamese. Additionally, we choose the same model as the one used by the BKEE [14] authors to ensure a fair comparison and demonstrate the improvements achieved by our enhanced dataset.

2.2. Data Augmentation

Two widely adopted methods for addressing data scarcity in event extraction are semi-supervised learning and distant supervision. Semi-supervised learning leverages a small amount of labeled data alongside a large pool of unlabeled data, using iterative learning techniques to improve model performance. In contrast, distant supervision automatically generates training labels by aligning raw text with external knowledge bases, enabling large-scale data annotation without manual labeling. While both methods aim to overcome the limitations of scarce annotated data, they differ in their assumptions, implementation strategies, and effectiveness in handling noisy labels.

Distant supervision offers an alternative way to automatically generate annotated training data by leveraging external knowledge bases such as Wikidata, Freebase, or domain-specific ontologies. This method works under the assumption that if an entity or trigger word in a sentence matches a known event in a structured database, the sentence is likely to express that event. While this approach significantly reduces the need for manual annotation and allows the creation of large-scale training datasets, it has its own set of challenges. The primary issue is label noise—many sentences that contain relevant

entities do not necessarily express the intended event. For example, the presence of the word "arrest" in a sentence does not always indicate an "Arrest" event, as it could be used in a different context (e.g., "The medicine helped arrest the disease"). We briefly discuss some popular approaches in applying Distant supervision:

- **Relation extraction:** Mintz *et al.* [12] presents distant supervision as a technique for relation extraction by aligning Freebase knowledge with text. The key assumption is that if two entities are known to have a relation in Freebase, any sentence containing them is a positive example of that relation. While this approach allows large-scale data generation without human annotation, it can be noisy in some cases because not all sentences mentioning an entity pair express the relation.
- **Open-Domain Event detection:** Instead of using a predefined knowledge base (like Freebase or Wikipedia infoboxes), Akari *et al.* [2] automatically generate training data from WordNet, using word sense disambiguation (WSD) and lexical heuristics to detect event triggers (verbs, nouns, adjectives, phrases). They further improve training data quality using Wikipedia for event phrases and a bidirectional LSTM (BLSTM) classifier with self-attention to classify event-related words. The BLSTM model trained on distant supervision data outperforms supervised models in both in-domain and out-of-domain settings, showing its robustness.

Semi-supervised learning, on the other hand, combines a small labeled dataset with a much larger unlabeled dataset to enhance model performance. The process typically starts with training a model on the available labeled data. Once trained, the model is used to generate pseudo-labels for the unlabeled data, which are then iteratively refined and incorporated into the training set. This iterative process allows the model to learn from a broader range of examples, gradually improving its performance. However, a key challenge in semi-supervised learning is maintaining the quality of pseudo-labels—low-quality labels can introduce noise into the dataset, potentially leading to error accumulation and reduced model accuracy. To address this, confidence-based filtering and consistency training are often used to ensure that only high-confidence predictions contribute to the learning process. We show some common approaches in applying Semi-supervised learning below:

- **Latent Representation-Based Event Induction:** Huang *et al.* [9] propose a Semi-Supervised Vector Quantized Variational Autoencoder (SS-VQ-VAE), which learns discrete latent type representations for both seen and unseen event types.

The model consists of a BERT-based encoder that extracts contextual trigger representations, a Vector Quantization (VQ) module that assigns triggers to discrete event type clusters, and a Variational Autoencoder (VAE) that helps regularize the type assignments by ensuring the reconstruction of event mentions. By integrating both labeled and unlabeled data, this approach enables the discovery of new event types while maintaining strong performance in supervised event detection tasks.

- **Self-Training:** Ferguson *et al.* [8] presents a semi-supervised self-training method for event extraction, leveraging paraphrase clusters found in news articles to improve event trigger detection. Traditional event extraction relies heavily on manually annotated datasets, which are both expensive and biased toward frequent event types. The authors propose an alternative approach: first, they automatically cluster sentences describing the same event from different news sources. Then, they apply a pre-trained event extraction system to detect high-confidence event mentions in these clusters. Once an event is identified in one sentence, the system propagates the label to the remaining sentences in the cluster, using word embedding similarity to assign event triggers.

Applied in our approach: Our model employs a self-training approach, where it first predicts event labels for plain sentences from articles on the same topic to generate pseudo-labeled data. A filtering mechanism is then applied based on the frequency of event occurrences and their associated entities. Only the qualified data is added to the corpus, which is subsequently used to retrain the model, improving its performance.

Several approaches leverage key components of sentences, such as entities, event triggers, to augment the data by generating diverse sentence variations. This not only enhances data diversity but also improves model generalization and robustness. Below, we explore some of the most widely used techniques:

- **Back-translation:** Xie *et al.* [22] introduces an approach to semi-supervised learning by leveraging advanced data augmentation techniques to enforce consistency in model predictions. The idea is to employ state-of-the-art augmentation techniques that have been effective in supervised learning, such as RandAugment, back-translation, and TF-IDF-based word replacement.
- **Synonym Replacement:** The study [5] introduces four augmentation methods: Label-wise Token Replacement (LwTR), Synonym Replacement (SR), Mention Replacement (MR), and Shuffle within Segments (SiS). LwTR replaces words with

other tokens from the dataset that share the same entity label, preserving label integrity. SR utilizes WordNet to substitute words with their synonyms, allowing slight semantic variation while maintaining readability. MR randomly replaces entire named entity mentions with others of the same entity type from the dataset, helping generalize entity patterns. Finally, SiS shuffles words within labeled segments while keeping the label order intact, promoting robustness to word order variations.

- **BERT:** The approach consists of two main steps: argument replacement and adjunct token rewriting [23]. Argument replacement swaps event arguments with similar entities that have played the same role in other sentences, adjunct token rewriting then refines the augmented sentences by replacing non-essential words (excluding triggers and arguments) using a fine-tuned BERT masked language model.
- **Schema-based:** This method was proposed by Jin et al. [10]. The approach consists of three key steps: schema graph sub-sampling, schema subgraph instantiation, and graph-to-text generation. It operates in three key steps: schema graph sub-sampling, where a connected subgraph is extracted from a predefined schema of continuous relevant sentences to represent a coherent event sequence; schema subgraph instantiation, where abstract event and entity types are replaced with real-world mentions from annotated datasets or external knowledge bases like ConceptNet; and graph-to-text generation, where the instantiated subgraph is converted into natural language using GPT-3.5.

Applied in our approach: We also utilize the schema of each annotated sentence to generate new sentences. Since there are no continuously relevant sentences, we extract the schema from individual sentences and then modify it by dividing it into sub-schemas or adding arguments such as *Time* or *Place* to enhance its diversity.

Chapter 3: Materials

This chapter provides a detailed explanation of the method used to automatically generate high-quality annotated data. The process consists of three main stages: *Data preparation*, *Silver corpus Formation*, and *Schema-based Data Augmentation*. These stages are visually represented in Figure 3.1. The stage of preparing the data is discussed in 3.1. The process of forming the Silver corpus is showed clearly in **Section 3.2**, and the data augmentation method used for enhancing the diversity of data will be described in **Section 3.3**

3.1. Data Preparation

The training set of BKEE [14] will be treated as the Golden corpus, which is used to train the Baseline model. Unlabeled data is collected from various Vietnamese news media sources. However, to effectively select candidates for the Silver corpus, the data needs to be grouped so that each group contains articles discussing the same topic. To achieve this, we identified two websites that continuously aggregate daily news from multiple online sources: *Báo Mới*¹ and *Google News*². These two sources already categorize articles by topic, making it easier to collect and organize related content. We gathered articles from these sites and grouped them by topic before breaking them down into individual sentences, forming multiple sets of raw sentences. In total, we collected approximately 72,000 articles, which were categorized into over 300 topic-based groups, containing a total of 2,673,796 unlabeled sentences. To generate annotated data, each of these sets is processed sequentially using the Baseline model, the model’s output consists of pseudo-labeled sentence sets corresponding to the original set of raw sentences, which serve as the foundation for further refinement and filtering in the data annotation pipeline.

3.2. Silver Corpus Formation

When an event occurs, there are often multiple parallel descriptions of that event available across articles due to the large number of different news sources. Taking the arti-

¹<https://baomoi.com>

²<https://news.google.com/home?hl=vi&gl=VN>

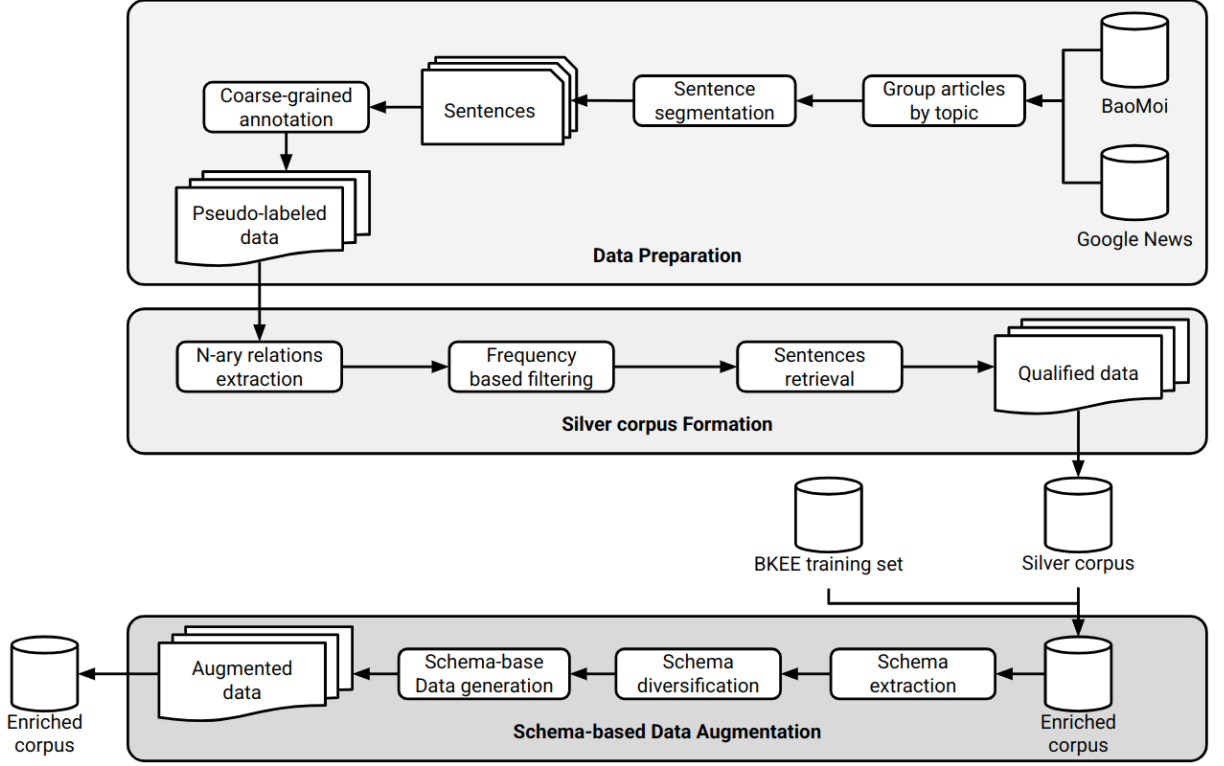


Figure 3.1: The process of enriching data

cles in the topic *THUẾ THU NHẬP CÁ NHÂN* categorized by *Báo Mới* as an example. Across multiple articles describing that topic, there will be several sentences mentioning the same event, and I take 3 sentences in 3 different articles as an example, showed in Figure 3.2. In those sentences, the event type *Meet* will be detected based on the word *hop* (meet). However, since the model can predict wrongly, so an event type may occur frequently, but the sentences are not relevant. Therefore, I leverage the N-ary relation of the Event Trigger with the associated entities to identify high-quality sentences. Taking the first sentence in Figure 3.2 as an example, the detected event trigger is *hop* (meet), and a binary relationship between it and its associated entities can be established. *Ngày 12/02/2025* (February 12, 2025) is an entity of type *TIME*, while *Bộ Tư pháp* (Ministry of Justice) is an entity of type *ORG*. They play the roles of *Time* and *Entity*, respectively, as illustrated in Figure 3.3. If multiple articles discuss the same topic, there will be several sentences across these articles that contain similar N-ary relations. Consequently, for the three sentences in Figure 3.2, three corresponding binary relations can be extracted. If a specific binary relation appears frequently across different sentences, surpassing a predefined threshold θ , then sentences containing that relation will be considered high-quality and selected for further processing.

For a better understanding, Figure 3.4 illustrates the N-ary relations that can be

- 1) Ngày 12/02/2025, Bộ Tư pháp đã tổ chức họp thẩm định đề nghị xây dựng Luật thuế TNCN. (*On February 12, 2025, the Ministry of Justice held an appraisal meeting on the proposal to draft the Personal Income Tax Law.*)
- 2) Trên cơ sở đó, ngày 12/2 Bộ Tư pháp đã tổ chức họp thẩm định đề nghị xây dựng Luật thuế thu nhập cá nhân. (*On that basis, on February 12, the Ministry of Justice held an appraisal meeting on the proposal to draft the Personal Income Tax Law.*)
- 3) Ngày 12/2/2025, Bộ Tư pháp đã tổ chức họp thẩm định với mục đích đề nghị xây dựng Luật Thuế thu nhập cá nhân. (*On February 12, 2025, the Ministry of Justice held an appraisal meeting with the aim of proposing the drafting of the Personal Income Tax Law.*)

Figure 3.2: Same meaning sentences across articles

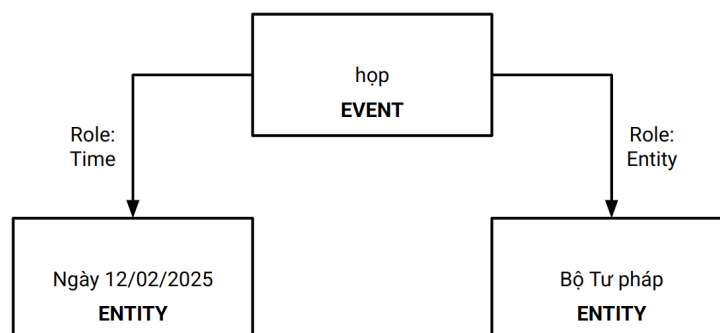


Figure 3.3: An example of N-ary relation between an Event trigger and associated entities

extracted across multiple articles discussing the same main topic. In this approach, relations are extracted from pseudo-labeled data based on their frequency and consistency within a given topic group. For example, if the relations r_1 and r_n frequently appear in articles covering the same topic, and their respective occurrence counts α_1 and α_n exceed a predefined threshold, they are considered valid and selected for further processing. Once these high-confidence relations are identified, the next step is to filter pseudo-labeled sentences that contain at least one of these N-ary relations. The selected sentences are then incorporated into the Silver corpus. By ensuring that only frequently occurring and contextually consistent relations are included, this approach enhances the reliability of the Silver corpus, maintaining a balance between data quantity and quality while reducing potential noise from incorrect pseudo-labels.

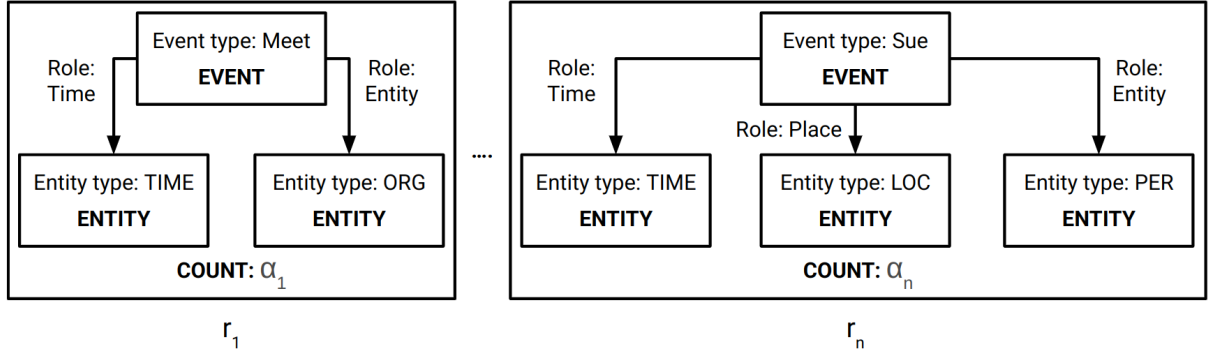


Figure 3.4: An example of N-ary relations across sentences in the same group

In summary, across N articles related to a topic, consisting of n sentences, m ($m \leq n$) sentences with event(s) detected, there should be M ($1 \leq M \leq 33$) event types extracted, and each event type can form K N-ary relations. If a relation is chosen, every sentence that forms the relation will be selected to be candidates for the Silver corpus. Given a group of related articles, a N-ary relation r_i between Event trigger type et_j and its associated entities, r_i is selected if

$$\begin{cases} \text{count}(|R_j|, m) \geq \mu \\ \text{count}(|r_i|, R_j) \geq \theta \end{cases}$$

$$\text{with } \theta = \begin{cases} 0, & \text{if } Q_3(C(UR_j)) - Q_1(C(UR_j)) \leq \frac{\max\{C(ur)|ur \in UR_j\}}{d} \\ \min\{C(ur)|ur \in UR_j\} + \frac{\max\{C(ur)|ur \in UR_j\} - \min\{C(ur)|ur \in UR_j\}}{2}, & \text{otherwise} \end{cases}$$

where R_j is the list of N-ary relations formed with et_j , UR_j is the set of unique relations in R_j ($UR_j = \{r|r \in R_j\}$), C_j is the count of $ur \in UR_j$ ($C_j : UR_j \rightarrow \{0, 1, 2, \dots\}$), Q_3 and Q_1 are *percentile₇₅* and *percentile₂₅*, respectively. μ and d are two pre-defined constants.

3.3. Schema-based Data Augmentation

In recent years, Large Language Models (LLMs) like ChatGPT and GPT-4 have achieved significant success in natural language processing (NLP) tasks and can also be used to generate augmented training data. However, relying solely on LLMs presents two potential issues [10]. First, simply masking text spans and filling in blanks does not provide enough new contexts for LLMs, leading to limited diversity in the generated data.

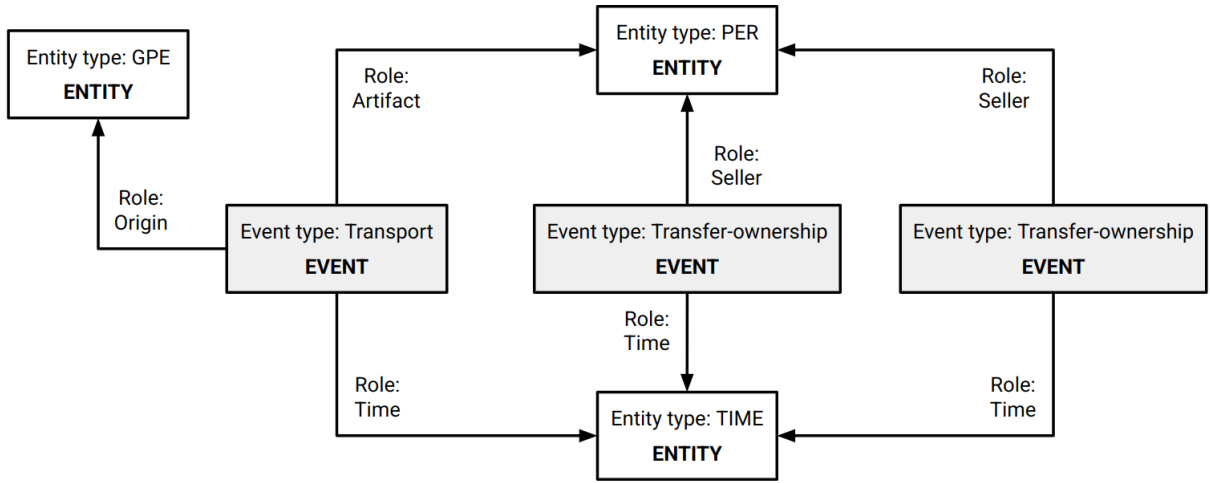


Figure 3.5: An example of the schema of a sentence

Moreover, generating sentences without proper constraints can result in low-quality sentences, which may negatively impact the training of event extraction models. Thus, as mentioned earlier, to increase data diversity, I applied a schema-based data augmentation method inspired by Jin et al. [10]. However, their approach relies on relationships between event triggers across multiple consecutive sentences. Finding such consecutive sentences that all contain events is a challenge because the original Golden corpus consists of individual sentences, and extracting continuous event-containing sentences from the pseudo-labeled data is nearly impossible. To address this issue, I propose an approach that only uses the schema of individual sentences to generate new ones. First, the schema of each sentence is extracted, forming a pool of schemas. Then, some schemas are randomly selected from this pool and used as input for a Large Language Model (LLM) to generate new sentences based on the selected schemas. The following paragraphs in this subsection will describe the schemas in detail and explain how they are used to generate new sentences.

A sentence can contain more than one event trigger. The schema is extracted based on the event triggers and their associated arguments. Let take the sentence in Figure 1.1 as an example, a schema can be extracted from that sentence, illustrated in Figure 3.5. As shown in the figure, the event schema captures patterns that can help regulate events and contextual information in newly generated synthetic data. I use schema graphs as event skeletons for data augmentation, ensuring that the generated sentences follow realistic event structures. The candidates for real event and entity mentions are gathered from the gold annotations from the Golden corpus and externally collected words. We want to create as many new contexts as possible. To do this, a new schema can be created

by dividing the original in different parts, but this can only be applied to sentences having $\text{count}(\text{event trigger}, s) \geq 2$. For sentences that have only one event trigger, a new schema can be created while still preserving the original context and maintaining sentence quality. This can be done by adding arguments such as *Time* or *Place* if they are missing from the original set of arguments. These arguments do not negatively affect the sentence because every event naturally takes place at a specific time and location. By adding arguments such as *Time* or *Place*, we ensure that the generated sentences remain contextually relevant and realistic. For example, if a sentence originally states that a company announced a new product, adding a time reference like "on Monday" or a place reference like "at a press conference in Hanoi" enhances the sentence without altering its meaning. This additional information provides more context for the event while maintaining grammatical correctness and logical coherence. This approach helps enrich the data without altering the core meaning of the sentence.

Once the schema is structured as a graph, the next step is to instantiate the entities and event trigger(s). At this stage, the nodes in the graph only represent abstract types rather than actual event or entity information. To assign real values to these nodes, we first create a mapping, denoted as M , which links event types to possible event mentions and entity types to potential entity candidates. The selection of event triggers and entity mentions is based on two sources: the *Internal* source and the *External* source. The *Internal* source consists of event mentions and entity names found directly in the labeled dataset, while the *External* source includes additional candidates retrieved externally. By incorporating both sources into a corpus of instances, we ensure a diverse and comprehensive set of instances for generating augmented sentences. A node (event trigger or entity) can be instantiated by randomly selecting an entity or event from that corpus of instances.

After the schema is instantiated, it is then converted into a JSON format, which serves as the input for generating new sentences. The schema is structured as a list of JSON objects, with each object corresponding to an event trigger and its associated entities. If the schema contains only one event, the list consists of a single element. In cases where multiple events exist within the schema, their order is randomized to increase the diversity of the generated sentences. This list of JSON objects is then used as input for GPT-4o. Once the sentence is generated, it is automatically annotated by matching the words in the text with those in the schema. Each matched word or phrase is then labeled with its corresponding event type or entity type, ensuring consistency in the dataset.

Chapter 4: Methods

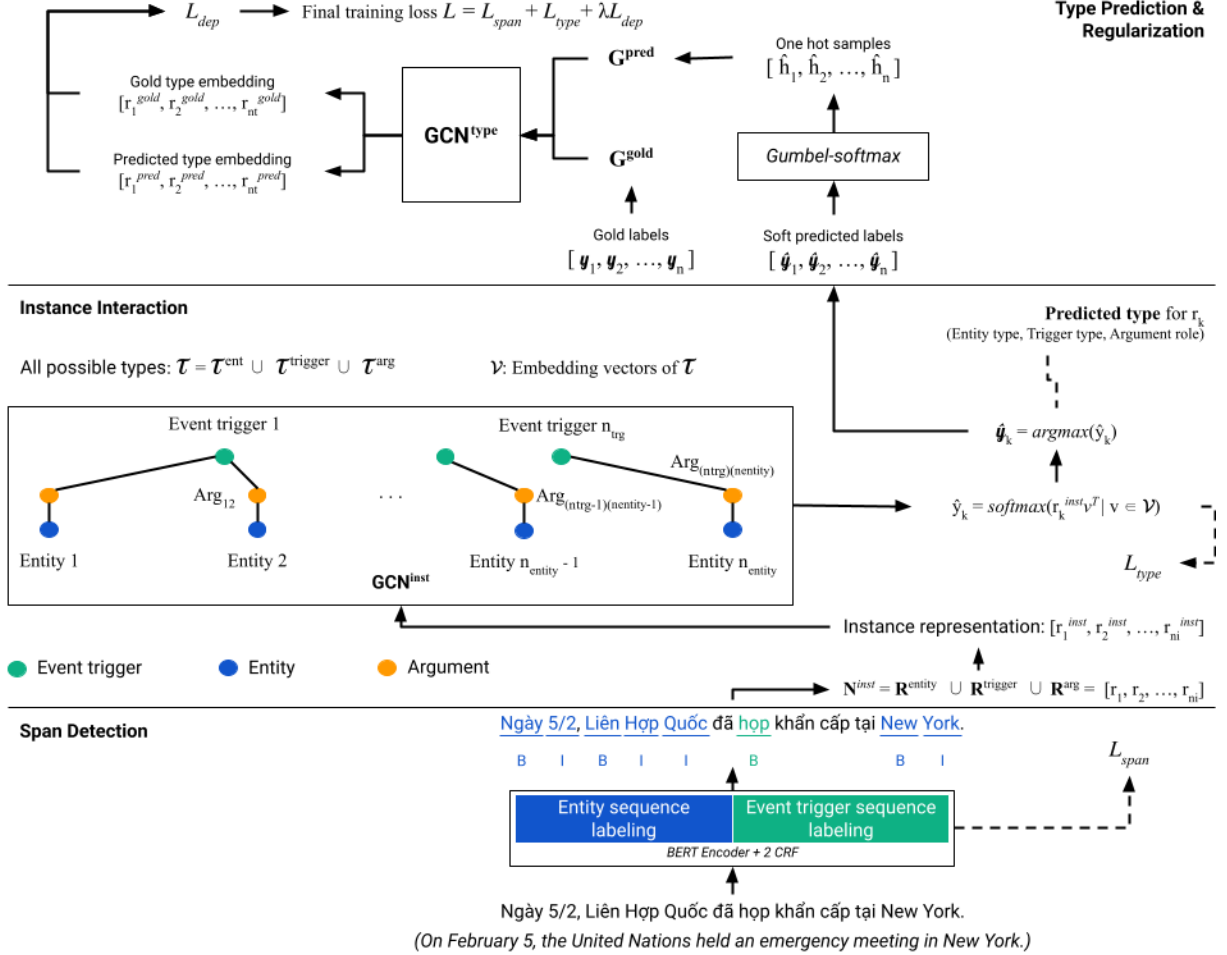


Figure 4.1: Overall architecture of the model

To effectively train the model with the generated data, we use a novel deep learning approach for joint information extraction (IE), called FourIE [13], designed to simultaneously perform three key IE tasks: EMD, ED, and EAE. Given an input sentence $\mathbf{w} = [w_1, w_2, \dots, w_n]$ (n words), unlike traditional methods that handle these tasks independently, this method leverages graph-based representation learning to capture cross-task interactions and label dependencies. There are two notable innovations: (1) an instance interaction graph that enriches representations by connecting related instances across tasks, and (2) a type dependency graph that encodes label constraints and enhances model consistency. The architecture of the model comprises three main stages: **Span Detection**, **Instance Interaction**, and **Type Dependency-based Regularization**, clearly showed in Figure 4.1. Each stage will be discussed in the following sections.

4.1. Span Detection

Span Detection is the initial step of the model. This component is responsible for identifying entity mentions and event triggers within an input sentence, which later serve as nodes in the interaction graph used for downstream tasks [13]. The problem is framed as a sequence labeling task, where each word in the sentence is tagged using the BIO tagging scheme (Begin, Inside, Outside). Unlike traditional Named Entity Recognition (NER) approaches, this step does not classify the extracted spans into specific entity or event types yet, leading to only three possible values: B, I, and O for the tags of the words. First, the input $\mathbf{w}$ is processed through the BERT [6] or XLM-RoBERTa [4] encoder to generate a sequence of vectors $X = [x_1, x_2, \dots, x_n]$, which serve as representations of $\mathbf{w}$. This vector sequence is then passed through two CRF layers to predict the optimal BIO tag sequences for identifying event mentions and event triggers. To train the span detection component, the model optimizes the negative log-likelihood losses, represented as $L_{\text{span}}^{\text{entity}}$ for entity mentions and $L_{\text{span}}^{\text{trigger}}$ for event triggers.

4.2. Instance Interaction

Instance Interaction is the second stage, after two separate span sets (entity and event triggers) have been extracted from input $\mathbf{w}$. This stage aims to capture and enhance interactions between instances across the tasks to improve prediction accuracy [13]. We first compute a representation vector for each span i, j ($1 \leq i \leq j \leq n$) in these two sets using the representation vectors $x_i, \dots, x_j$. Let define $R^{\text{entity}} = \{e_1, e_2, \dots, e_{n_{\text{entity}}}\}$ ($n_{\text{ent}} = |R^{\text{entity}}|$) and $R^{\text{trigger}} = \{t_1, t_2, \dots, t_{n_{\text{trigger}}}\}$ ($n_{\text{trigger}} = |R^{\text{trigger}}|$) are the sets of span representation vectors for the entities and event triggers in $\mathbf{w}$. The use of these two sets of span representation vectors will be described in the following two sub-sections.

4.2.1. Instance Representation

Once R^{entity} and R^{trigger} are formed, we define and construct vector representations for instances in the three IE tasks: EMD, ED, and EAE. These vector representations serve as inputs for the interaction graph. A prediction instance refers to an element within a sentence that the model classifies into a specific type, such as an Entity, Event trigger, or Argument. Consequently, Entity instances and Event trigger instances are directly derived from R^{entity} and R^{trigger} , respectively, and are used as their initial representation vectors. However, predicting an Argument requires both an event trigger and an entity. Therefore,

argument instances are defined as $R^{\text{argument}} = \{arg_{ij} = [t_i, e_j] \mid t_i \in R^{\text{trigger}}, e_j \in R^{\text{entity}}\}$. Based on this, the initial representation vectors for argument instances are generated accordingly.

4.2.2. Instance Interaction

To establish interactions between related prediction instances, we construct a graph that links the three types of prediction instances [13]. This graph, denoted as G^{inst} , consists of a set of nodes N^{inst} and a set of edges E^{inst} . The node set is formed by combining the three sets of prediction instances, meaning $N^{\text{inst}} = R^{\text{entity}} \cup R^{\text{trigger}} \cup R^{\text{argument}}$. Each edge in E^{inst} represents a connection between two related prediction instances. Specifically, an entity instance node e_i is linked to all argument instance nodes $arg_{ij} = [t_j, e_i]$, and an event trigger node t_j is also connected to these argument instance nodes. This structure enables effective information sharing between related instances in the model.

This graph is then consumed by a Graph Convolutional Network (GCN) model in order to enrich the representation vector for prediction instances. The initial representation vectors of all instances nodes in N^{inst} are defined as $\{r_1, r_2, \dots, r_{n_i}\}$. The adjacency matrix A^{inst} is then created, with $N_{ij}^{\text{inst}} = 1$ means nodes r_i and r_j are connected. Then the enriched interaction representation vectors r^{inst} are then computed by the GCN model as: $r_1^{\text{inst}}, r_2^{\text{inst}}, \dots, r_{n_i}^{\text{inst}} = GCN(A^{\text{inst}}; r_1, r_2, \dots, r_{n_i}; N_i)$, with N_i as the number of layers.

Finally, the vectors $r_1^{\text{inst}}, r_2^{\text{inst}}, \dots, r_{n_i}^{\text{inst}}$ are utilized to perform the three tasks: EMD, ED, and EAE. We define $\mathcal{T} = \mathcal{T}^{\text{entity}} \cup \mathcal{T}^{\text{trigger}} \cup \mathcal{T}^{\text{argument}}$, where $\mathcal{T}^{\text{entity}}$ represents the possible entity types, and similarly for event triggers and arguments. We define t_k as the task index ($t_k \in \{\text{entity}, \text{trigger}, \text{argument}\}$), and y_k denote the ground-truth type of t_k . Each type in $\mathcal{T}$ is associated with an embedding vector v ; hence, the embedding vectors for $\mathcal{T}$ are defined as $\mathcal{V}$, and for $\mathcal{T}^{t_k}$, they are represented as $\mathcal{V}^{t_k}$, which is randomly initialized. To predict the type of an instance $r_k \in N^{\text{inst}}$, we compute the dot product between r_k^{inst} and each embedding vector in $\mathcal{T}^{t_k}$, which determines the likelihood that r_k belongs to a specific type in $\mathcal{T}^{t_k}$. The probability distribution is then given by:

$$\hat{y}_k = \text{softmax}(r_k^{\text{inst}} v^T \mid v \in \mathcal{V}^{t_k})$$

Finally, the predicted type $\hat{y}_k$ for r_k is obtained by selecting the type with the highest probability:

$$\hat{y}_k = \text{argmax}(\hat{y}_k)$$

4.3. Type Prediction And Regularization

The goal of this regularization is to model the global type dependencies across the three IE tasks (Entity Mention Detection, Event Detection, Event Argument Extraction). This dependency modeling helps refine the representation vectors and improve overall prediction accuracy.

First of all, two dependency graphs G^{gold} and G^{pred} are constructed using the golden types $y = y_1, y_2, \dots, y_{n_i}$ and predicted types $\hat{y} = \hat{y}_1, \hat{y}_2, \dots, \hat{y}_{n_i}$, to capture the global type dependencies. The nodes represent types in $\mathcal{T}$, and edges define relationships between types (e.g. Entity-Argument Edges: Link argument role types to corresponding entity types). A^{gold} and A^{pred} are adjacency matrices for G^{gold} and G^{pred} , respectively. The representation vectors for the dependency graphs will then be computed using GCN with initial type embedding $\mathcal{T} = [v_1, \dots, v_{n_t}]$: $r_1^{gold}, r_2^{gold}, \dots, r_{n_t}^{gold} = GCN(A^{gold}; v_1, \dots, v_{n_t}; N_t)$, and the same goes to predicted representation vectors. The overall loss function here is $L_{dep} = \sum ||r_i^{gold} - r_i^{pred}||_2^2$.

Since G^{pred} is based on discrete predictions, backpropagation is not straightforward. To address this, A^{pred} is approximated with a differentiable function $\hat{A}^{pred}$, the $\hat{A}^{pred}$ is computed by:

$$\hat{A}^{pred} = \sum_{(i,j) \in I^{inst}} \exp(-\beta (B - \text{int}_t - j)^2)$$

with $I^{inst} = \{(i, j) \mid A_{ij}^{pred} = 1\}$, matrix $B = \{b_{ij}\}_{i,j=1,\dots,n_t}$, and β is a large constant. Additionally, since type predictions are categorical, the model employs Gumbel-Softmax, a reparameterization trick that allows gradient-based optimization by approximating one-hot categorical distributions with continuous relaxations. The final loss function incorporates this regularization term alongside standard span detection and type prediction losses: $L = L_{span}^{entity} + L_{span}^{trigger} + L_{type} + \lambda L_{dep}$, where λ is a trade-off parameter. By enforcing structural consistency across tasks, this method significantly enhances the quality of joint IE, outperforming prior models across multiple datasets and languages.

Chapter 5: Experiments and Results

5.1. Evaluation Metrics and Implementation

In this section, we evaluate the performance state-of-the-art Event Extraction (EE) models on the new dataset to highlight the inherent challenges it presents.

5.1.1. Training Methods and Evaluation Metrics

In our experiments, we use the joint model with different data setups to evaluate the impact of each stage in the study. Specifically, we train the model using: (1) the plain BKEE training set, (2) the enriched corpus, which combines the BKEE training set and the Silver corpus, (3) the enriched corpus without augmented data, and (4) the BKEE training set with the augmented data. For training with the enriched corpus, we employ three approaches: (a) training the joint model from scratch, (b) first creating a small corpus by combining the BKEE training set with approximately one-fifth of the Silver corpus to build a baseline model. This baseline model is then used as a pre-trained model and further trained on the enriched corpus. (c) The model is further trained using the baseline model, which was trained on the BKEE training set, as the pre-trained model. For convenience, we refer to the pre-trained model trained on the BKEE training set as PM_1 , while the other model is denoted as PM_2 . In all approaches, each task — EMD, ED, and EAE — is evaluated on the original BKEE testing set using the F1 score.

5.1.2. Implementation and Configurations

Our model was implemented using Python 3.10.16, PyTorch 2.0.1+cu117, and the Transformers library version 4.47.1. The training was conducted on an NVIDIA GeForce RTX 2080 Ti GPU with 12GB of VRAM. For the pre-trained models mentioned, we utilize models and toolkits available under the public-sharing policy of the Hugging Face Model Hub and GitHub, which are:

- XLM-RoBERTa [4] model FacebookAI/xlm-roberta-large for encoding.
- VnCoreNLP toolkit [18] for word tokenization.

The configuration variables of our model are as follows: the linear dropout rate is set to 0.4, the linear activation function is sigmoid, the number of hidden layers N_i is 2, $\beta = 1000$, $\lambda = 0.5$, and the learning rate is $5e-6$. For all approaches, the maximum number of epochs is set to 25. Due to resource limitations, the batch size is fixed at 1.

5.2. Evaluation Results

5.2.1. Evaluation Results

Table 5.1 shows the performances by F1 score of our approach and the original approach. Our approach, which trains on the enriched corpus and leverage the pre-trained model, leads in the ranking evaluation metric in all three tasks EMD, EE and EAE. The results of the three EE tasks indicate that training the model with a pre-trained model on the enriched corpus is an effective approach for addressing all three tasks.

Approach	EMD	ED	EAE
Training BKEE training set	55.6	59.3	51.1
Training enriched corpus with pre-trained model PM_1	56.88	61.24	52.5
Training enriched corpus with pre-trained model PM_2	59.11	61.73	55.17
Training enriched corpus from scratch	57.43	62.22	53.09
Training enriched corpus without augmented data	56.59	60.33	51.32
Training BKEE training set with augmented data	57.46	60.83	52.80
<i>The highest results for each metric (EE task) are highlighted in bold.</i>			

Table 5.1: Results of approaches across three EE tasks

5.2.2. Contribution of Each Proposed Components

In this subsection, we analyze the impact of each proposed component, including pseudo-labeling with event occurrence frequency filtering, schema-based data augmentation. Each component plays a crucial role in improving event extraction performance, and their contributions are evaluated based on comparative results.

To address data scarcity, we applied a distant supervision method to generate pseudo-labeled data from unlabeled sentences. The filtering mechanism, based on event occurrence frequency and N-ary relations, significantly improved the quality of the Silver corpus by selecting only high-confidence instances. Training with the enriched corpus

without augmented data (a combination of the BKEE training set and sentences filtered by frequency) results in a higher F1 score across all three tasks compared to using the original BKEE training set.

To further improve data diversity, we implemented a schema-based data augmentation approach inspired by previous work but tailored for individual sentence schemas rather than multi-sentence structures. When training with the BKEE training set and augmented data, the model outperformed the baseline BKEE-only training approach, achieving 57.46 for EMD, 60.83 for ED, and 52.80 for EAE. The improvement over the baseline confirms that augmenting training data with diverse, structured variations enhances generalization and helps the model handle different sentence structures.

Notably, leveraging the enriched corpus—combining the BKEE training set and all new data—to continue training the baseline (pre-trained) model yields the best performance across all tasks, demonstrating that training with a pre-trained model is more efficient than training from scratch.

UETCorn-V2

| 0.470 **0.444** 0.339 0.435

5.3. Error Analysis

To improve the proposed model and incentivize future research, we have analyzed the output on the test set to find out problems that need to be taken into account. Some notable errors are shown in Table 5.2.

The first type of common errors is **Span error**: This type of error occurs in the EMD and ED task sometimes. These errors occur when the model captures part of a mention by does not overlap completely with the gold one, or the model over captures the span. For example, the entity *Ngân hàng Nông nghiệp và Phát triển nông thôn Việt Nam Agribank* (Vietnam Bank for Agriculture and Rural Development Agribank) is only partially detected as "Agribank", or *Bí thư tỉnh ủy và Chủ tịch Hội đồng nhân dân tỉnh Bến Tre Hồ Thị Hoàng Yến* (Secretary of the Provincial Party Committee and Chairwoman of the People’s Council of Ben Tre Province, Ho Thi Hoang Yen) should be recognized as two entities *Bí thư tỉnh ủy* and *Chủ tịch Hội đồng nhân dân tỉnh Bến Tre Hồ Thị Hoàng Yến*. The sentences containing these mentions are illustrated in Table 5.2, corresponding to sentence (1) and (2), respectively. This type of error often occurs with sentences that contains entity mentions, or event triggers with complex structure or too lengthy.

The second type is **Isolated Entity Detection Failure**: This type of error occurs in the EMD task, meaning that an entity appearing independently, without being linked to an event trigger, fails to be detected. This limitation arises because the model relies heavily on event triggers to recognize relevant entities. Take the third sentence in Table 5.2 as an example: ideally, two entities should be identified—*ông Hùng* (Mr. Hung) and *năm 2023* (the year 2023). However, in the absence of an event trigger, the model overlooks these entities, resulting in missed detections. This issue highlights the challenge of identifying standalone entities, especially in cases where contextual clues are insufficient to compensate for the lack of an explicit event mention.

The third type of error is **Polysemy**: Words with multiple meanings in different contexts can sometimes lead to misclassification. For example, in the fourth sentence of Table 5.2, the term "Jordan" could be incorrectly identified as PER (Person) instead of GPE (Geo-Political Entity) under the ACE2005 tagging scheme. Such errors arise when the model fails to distinguish between different contextual meanings of the same word. However, this type of misclassification is relatively uncommon compared to other entity recognition errors.

A type of error sometimes occurs in the EAE task, is **Ambiguous Context**: This type of error frequently occurs in the EAE task, where sentences containing events tend to be lengthy and packed with overlapping contextual elements, increasing the likelihood of misclassification. Due to the ambiguous context, the system may struggle to accurately assign event roles. For instance, in the fourth sentence in Table 5.2, the entity "An" was mistakenly labeled as the "Agent" of the "Transport" event. However, in reality, "An" is not the one being transported but merely maintaining communication, leading to a misinterpretation of role assignments within the event structure.

Sentence	Error
(1) <i>Lãnh đạo Ngân hàng Nông nghiệp và Phát triển nông thôn (Agribank) Chi nhánh Bắc Yên Bái trao tiền hỗ trợ gia đình chị Nông Thị Đông ở thị trấn Yên Bình.</i> (Leaders of the Agricultural Bank and Rural Development (Agribank) North Yen Bai Branch donated financial support to the family of Ms. Nong Thi Dong in Yen Binh Town)	Wrong Entity detected: Agribank
(2) <i>Bí thư tỉnh ủy và Chủ tịch Hội đồng nhân dân tỉnh Bến Tre Hồ Thị Hoàng Yến chủ trì phiên họp .</i> (Secretary of the Provincial Party Committee and Chairwoman of the People’s Council of Ben Tre Province, Ho Thi Hoang Yen, chaired the meeting)	Wrong Entity detected: Bí thư tỉnh ủy và Chủ tịch Hội đồng nhân dân tỉnh Bến Tre Hồ Thị Hoàng Yến
(3) <i>Trước đây , gia đình ông Hùng thuộc diện hộ nghèo , đến năm 2023 là hộ cận nghèo.</i> (In the past, Mr. Hung’s family was classified as a poor household, but by 2023, they became a near-poor household.)	No entity detected
(4) <i>Jordan bày tỏ lo ngại về những căng thẳng gia tăng trong khu vực.</i> (Jordan expressed concerns over the escalating tensions in the region)	Wrong Entity type detected: Jordan - PER
(5) <i>Hải đến nhà An, và An dùng điện thoại để giữ liên lạc với Hải trong quá trình di chuyển</i> (Hai went to An’s house, and An used his phone to stay in touch with Hai during the journey)	Wrong Argument detected: An is detected as <i>Agent</i> in the event type <i>Transport</i> (identified by the word <i>di chuyển</i>)

Table 5.2: Some notable errors on the test set output of the model

Conclusions

This report presents a study on an enhanced distant supervision method designed to address data scarcity in event extraction (EE) tasks. Additionally, we implement a joint model leveraging a Graph Convolutional Network (GNN) to capture dependencies between prediction instances, enabling the simultaneous execution of all three EE tasks. This approach eliminates the need for a pipeline of multiple distinct models, streamlining the process and improving efficiency.

The distant supervision method is designed to select high-quality candidates from the pseudo-labeled set by analyzing the occurrence frequency of N-ary relations between each event trigger and its associated entities. Since sentences from different articles on the same topic often convey similar meanings, certain patterns of N-ary relations tend to appear frequently across these articles. Subsequently, a schema-based data augmentation method is employed to diversify the dataset by generating new instances with varied contexts and schemas. This approach enhances the model’s ability to generalize effectively across different scenarios.

We implement a joint model with dependency-aware learning, leveraging Graph Convolutional Networks (GCNs) to effectively capture interdependencies between instances—specifically, entities, event triggers, and arguments. Our approach constructs an interaction graph to explicitly model cross-task dependencies, enabling the model to enrich instance representations by incorporating contextual information from related elements. Additionally, we adopt a type dependency graph that captures structural relationships between predicted labels, further enhancing consistency and robustness across the three core event extraction (EE) tasks: EMD, ED, EAE. This integration mitigates error propagation and improves the overall performance of joint information extraction.

The evaluation results demonstrate the strong potential of this approach. Training methods that utilize the full corpus or subsets of it consistently achieve higher F1 scores across all three EE tasks compared to training with the original BKEE training set. Specifically, the highest F1 score improvements for EMD, ED, and EAE are 3.51%, 2.92%, and 4.07%, respectively, over the baseline model. Additionally, the results indicate that each stage of the proposed method contributes positively to the model’s overall performance. An error analysis was conducted to identify existing shortcomings, which will be addressed in future work to further enhance performance.

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